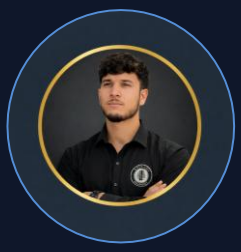

Miraj Ud Din

BS Artificial Intelligence · The University of Agriculture, Peshawar

PROFESSIONAL SUMMARY

Artificial Intelligence undergraduate (6th semester) with strong expertise in Machine Learning, Deep Learning, and Data Science. Proficient in Python, TensorFlow, Keras, and the full ML pipeline, from raw data preprocessing and feature engineering to hyperparameter optimization (Optuna / TPE) and Streamlit-based web deployment. Experienced in comparative model analysis across multiple regression and classification algorithms. Aspiring AI researcher with published work on smartphone price prediction using advanced ML techniques.

EDUCATION

BS Artificial Intelligence

The University of Agriculture, Peshawar | 2023 – 2027 (Expected)

Institute of Computer Sciences and Information Technology (ICS/IT)

CGPA: **3.65 / 4.00** | 6th Semester

PROJECTS

Smartphone Price Prediction – Comparative ML Regression Analysis

XGBoost · Optuna · Streamlit Deployment | Best Model R^2 = 0.9013

Built and evaluated 18 regression algorithms (Linear Regression, Ridge, Lasso, ElasticNet, KNN, SVR, Decision Tree, Random Forest, ExtraTrees, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost, and more) on an uncleaned Kaggle smartphone dataset. Applied Optuna TPE over 50 trials for automated hyperparameter tuning. XGBoost emerged as the best model (R^2 = 0.9013, MAE = 0.2102, MSE = 0.0948). Deployed as a real-time interactive Streamlit app with multi-currency output (PKR / USD / INR / CNY). The research paper is currently under submission to a journal.

Live App GitHub Repo

Credit Card Fraud Detection – ML Classification System

Logistic Regression · Random Forest · XGBoost · SMOTE

Developed a classification pipeline to detect fraudulent credit card transactions. Applied Logistic Regression, Random Forest, and XGBoost on a heavily imbalanced dataset; used SMOTE and appropriate preprocessing to handle class imbalance. Evaluated using accuracy, precision, recall, and F1-score and deployed as a live Streamlit web app for real-time fraud detection.

Live App GitHub Repo

PUBLICATIONS

A Comparative Analysis of Machine Learning Regression Models for Smartphone Price Prediction with Optuna Hyperparameter Optimization and Streamlit Deployment

Miraj Ud Din

Artificial Intelligence Engineer

Data Science | Machine Learning | Data Analysis | Deep Learning | AI Researcher

CONTACT

mirajuddin1357@gmail.com

Peshawar, KPK, Pakistan

+92 306 021 3131

Portfolio Website

Kaggle Profile

GitHub

PROGRAMMING

Python (Libraries)	95%
SQL	90%
HTML / CSS / JS	60%
C++	50%

TOOLS & PLATFORMS

NumPy, Pandas, Matplotlib, Seaborn, Plotly, Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, Keras, ANN, CNN, RNN, LSTM, NLTK, spaCy, Optuna, SMOTE, Git & GitHub, VS Code, Antigravity, Jupyter Notebook, Google Colab, Streamlit, Vercel, Firebase, Excel, Word, PowerPoint, EndNote

LANGUAGES

- English (Fluent)
- Urdu (Native)
- Pashto (Native)

LINKS

GitHub LinkedIn

Kaggle Portfolio

Under Review – Journal Submission 2026

Comprehensive comparative study evaluating 18 ML regression models with Optuna-based hyperparameter optimization. XGBoost was identified as the best model. Full IMRaD structure with 34 APA-cited references and a structured abstract.

CERTIFICATIONS

- **NAVTTC: Artificial Intelligence** (Machine Learning · Deep Learning · Communication) View Certificate
- **Microsoft Azure AI Fundamentals (AI-900)** View Certificate
- **DataCamp: Data Science for Business** View Certificate
- **Cisco: Introduction to Data Science** View Certificate
- **DataCamp:** Introduction to Python View Certificate
- **DataCamp:** Introduction to SQL View Certificate
- **DataCamp:** Introduction to Excel View Certificate
- **Typing.com:** Typing Proficiency View Certificate

RESEARCH INTERESTS

Artificial Intelligence Researcher, Machine Learning, Deep Learning, Data Science, Data Analysis, Optimization, Natural Language Processing

EXTRA ACTIVITIES

LetTech Ramzan Series 2026 – One-Week Bootcamp

Entrepreneurship · AI Tools for Students · AI in IT Operations · AI Cybersecurity & SDGs · Career Crafting · Solutions That Save Lives View Certificate

Career Direction Seminar View Certificate